TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	85.6 ± 2.46
Correlation Stability	89.6 ± 0.80
Distribution Consistency	88.2 ± 3.22
Entropy Preservation	90.2 ± 1.25
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.4 ± 0.66
Feature Accuracy	95.9 ± 0.94
Novelty Score	0.9 ± 0.02

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Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	41.903	+0.703	1.7%	OK
Sex	1.103	1.089	-0.015	1.3%	OK
Steroid	1.506	1.556	+0.050	3.3%	OK
Antivirals	1.845	1.855	+0.010	0.5%	OK
Fatigue	1.351	1.282	-0.068	5.1%	OK
Malaise	1.604	1.556	-0.047	3.0%	OK
Anorexia	1.792	1.774	-0.018	1.0%	OK
Liver Big	1.828	1.831	+0.003	0.2%	OK
Liver Firm	1.583	1.629	+0.046	2.9%	OK
Spleen Palpable	1.800	1.766	-0.034	1.9%	OK
Spiders	1.660	1.669	+0.009	0.6%	OK
Ascites	1.867	1.831	-0.036	1.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	85.6 ± 2.46
Correlation Stability (*2)	89.6 ± 0.80
Distribution Consistency (*3)	88.2 ± 3.22
Entropy Preservation (*4)	90.2 ± 1.25
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.4 ± 0.66
Feature Accuracy (*10)	95.9 ± 0.94
Novelty Score (*11)	0.9 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.8258064516129032	0.8548387096774193	0.016129032258064512	
F1	0.5132478632478632	0.4781385281385281	0.5807401422107304	0.06749227896286719	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	0.8258064516129032	0.864516129032258	0.01290322580645159	
F1	0.5385593354014405	0.4592857142857124	0.6114608920491273	0.07290155664768669	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	0.832258064516129	0.8451612903225806	0.006451612903225812	
F1	0.5718392134181608	0.4517521367521367	0.4967099567099567	-0.07512925670820408	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8129032258064516	0.7741935483870968	0.8193548387096774	0.006451612903225845	
F1	0.5256565656565656	0.3968283024165377	0.4459013209013208	-0.07975524475524479	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	0.7645161290322581	0.832258064516129	0.04193548387096774	
F1	0.561449871078354	0.5453299916457811	0.6026495726495725	0.04119970157121861	

Model: GB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8483870967741934	0.8096774193548386	0.8612903225806452	0.012903225806451613	
F1	0.5644355644355643	0.4545992895992896	0.5689377289377289	0.004502164502164513	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7838709677419355	0.7580645161290323	0.7516129032258065	-0.032258064516129024	
F1	0.4456177156177157	0.4232822148611624	0.3955535967300673	-0.05006411888764826	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.861290322580645	0.832258064516129	0.8580645161290322	-0.0032258064516129	
F1	0.6001225898284723	0.4088034188034187 5	0.5668956043956045	-0.03322698543286779	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	0.8032258064516128	0.8064516129032258	0	
F1	N/A	0.0	N/A	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	0.7967741935483871	0.7838709677419355	0.0032258064516129	
F1	N/A	0.1126068376068376 1	0.0403846153846153 9	0.04038461538461539	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	54.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0
2	42.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
3	38.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
4	45.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0
5	50.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	49.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	32.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
2	46.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	40.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0
4	36.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
5	53.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	70.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	25.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
2	65.0	1.0	2.0	2.0	1.0	2.0	2.0	1.0	1.0	2.0
3	57.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	38.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	2.0	2.0
5	64.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	50.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	36.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	27.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	33.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	36.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
5	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	30.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0	2.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	40.0	1.0	2.0	2.0	2.0	2.0	1.0	1.0	1.0	1.0
2	52.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
3	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	36.0	1.0	1.0	2.0	2.0	2.0	2.0	1.0	1.0	2.0
5	56.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
6	50.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
2	44.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
3	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
4	48.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
5	38.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
6	51.0	2.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	34.0	1.0	1.0	1.0	1.0	2.0	2.0	1.0	1.0	2.0
2	37.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
3	33.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
4	56.0	1.0	1.0	1.0	1.0	2.0	2.0	1.0	1.0	2.0
5	33.0	1.0	1.0	1.0	2.0	2.0	1.0	2.0	1.0	2.0
6	30.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	42.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
2	38.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
3	31.0	2.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
4	25.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
5	45.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
6	47.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	64.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
2	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0
3	48.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
4	49.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
5	42.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
6	45.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
2	30.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
3	34.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0
4	32.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
5	35.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
6	51.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	1.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0